

Gypsum Pulse Injection Example

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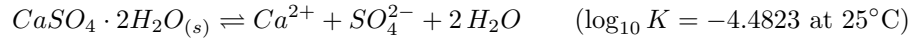
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1 Chemistry

We consider a reactive transport problem in a one-dimensional column initially containing gypsum ($CaSO_4 \cdot 2H_2O$) in equilibrium with resident water of low salinity. A high-concentration calcium-chloride pulse (equivalent to dissolved $CaCl_2$) is imposed as an initial condition in one interior cell of the column, while clean water in equilibrium with gypsum enters through the inflow boundary. This problem tests the ability of the Water Mixing Approach (WMA) to transport reactive solutes in a system governed purely by equilibrium reactions, with no kinetic reactions. The aqueous species considered are Ca^{2+} , Cl^- , SO_4^{2-} , and H_2O . The mineral is gypsum ($CaSO_4 \cdot 2H_2O$), which is present as a pure solid phase with constant activity equal to one. Throughout this document we assume ideal activities for the aqueous species ($\gamma_i = 1$), so that activities and molal concentrations coincide ($a_i = c_i$).

We consider the following equilibrium reaction:

- Gypsum dissolution/precipitation:



The equilibrium constant $\log_{10} K = -4.4823$ is taken from the thermodynamic database file `master25_modif.dat` used by the simulation code, which follows the standard convention of treating H_2O as constant activity ($a_{H_2O} = 1$) and absorbing it into K ; equivalently, the mass-action expression for gypsum reads $K = a_{Ca^{2+}} a_{SO_4^{2-}}$. There are no kinetic reactions in this problem ($\mathbf{S}_K = \mathbf{0}$). Since $CaSO_4 \cdot 2H_2O_{(s)}$ and H_2O both have constant activity, the variable activity species are Ca^{2+} , Cl^- , and SO_4^{2-} .

Both in the domain (column interior) and at the inflow boundary, the water is in equilibrium with gypsum, so only the gypsum reaction applies in both cases. Consequently $\mathbf{S}_{e,v}^{dom} = \mathbf{S}_{e,v}^{inf}$, and the equilibrium stoichiometric matrix, restricted to variable activity species, is

$$\mathbf{S}_{e,v} = \begin{pmatrix} Ca^{2+} & Cl^- & SO_4^{2-} \\ 1 & 0 & 1 \end{pmatrix},$$

where the single row corresponds to gypsum dissolution/precipitation: producing one Ca^{2+} , leaving Cl^- unaffected, and producing one SO_4^{2-} .

1.1 Component matrix

Following De Simoni et al. (2005), the component matrix $\mathbf{U}$ is defined such that $\mathbf{U}\mathbf{S}_{e,v}^T = \mathbf{0}$, ensuring that the components $\mathbf{u} = \mathbf{U}\mathbf{c}$ are conservative quantities not affected by equilibrium reactions. Here $\mathbf{S}_{e,v}$ denotes the equilibrium stoichiometric matrix restricted to variable activity species (i.e. excluding constant activity species), which in this problem are H_2O and the mineral phase $CaSO_4 \cdot 2H_2O_{(s)}$.

Splitting $\mathbf{S}_{e,v} = (\mathbf{S}'_{e,v} \mid \mathbf{S}''_{e,v})$ into blocks corresponding to primary and secondary variable activity species, and redefining so that $\mathbf{S}''_{e,v} = -\mathbf{I}$, the component matrix takes the form

$$\mathbf{U} = \left(\mathbf{I}_{N_u} \mid \mathbf{S}'_{e,v}^T \right), \quad (1.1)$$

where $N_u = N_v - N_r$ is the number of components, N_v is the number of variable activity species and N_r the number of equilibrium reactions.

In the domain, we have $N_v = 3$ variable activity species (Ca^{2+} , Cl^- , SO_4^{2-}) and $N_r = 1$ equilibrium reaction (gypsum dissolution/precipitation), giving $N_u = 3 - 1 = 2$ components. We choose SO_4^{2-} as the secondary species (it has coefficient +1 in the gypsum dissolution/precipitation reaction), so the primary species are Ca^{2+} and Cl^- .

The secondary block is $\mathbf{S}_{e,v}''^{dom} = (1)$, which we redefine to $-\mathbf{I}$ by multiplying the gypsum row by -1 . This yields the transformed primary block $\mathbf{S}_{e,v}'^{dom} = (-1, 0)$ (coefficients of Ca^{2+} and Cl^- after the sign change). The component matrix is therefore $\mathbf{U}^{dom} \in \mathbb{R}^{2 \times 3}$:

$$\mathbf{U}^{dom} = (\mathbf{I} \mid \mathbf{U}_2^{dom})$$

where, with rows corresponding to the primary species Ca^{2+} and Cl^- ,

$$\mathbf{U}_2^{dom} = (\mathbf{S}_{e,v}'^{dom})^T = \begin{pmatrix} SO_4^{2-} \\ -1 \\ 0 \end{pmatrix}.$$

Written out explicitly in the natural species order,

$$\mathbf{U}^{dom} = \begin{pmatrix} Ca^{2+} & Cl^- & SO_4^{2-} \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix},$$

which yields the conservative components

$$\begin{aligned} u_1^{dom} &= c_{Ca^{2+}} - c_{SO_4^{2-}}, \\ u_2^{dom} &= c_{Cl^-}. \end{aligned}$$

The physical interpretation is straightforward: gypsum dissolution/precipitation produces equal moles of Ca^{2+} and SO_4^{2-} , so their difference u_1 is invariant under this reaction; and the reaction does not affect Cl^- , so u_2 is also unaffected by it. Both components are still subject to advection and dispersion — they are conservative *with respect to the chemical reactions*, not constant in space and time.

Component matrix of the boundary (inflow) water. At the inflow, the same gypsum equilibrium applies and no additional reactions are active, so the set of variable activity species, the equilibrium stoichiometric matrix and the choice of primary/secondary species are all identical to those of the domain. Consequently the chemistry of the inflow water is identical to that of the domain water, and the boundary component matrix coincides with the domain one:

$$\mathbf{U}^{inf} = \mathbf{U}^{dom} = \begin{pmatrix} Ca^{2+} & Cl^- & SO_4^{2-} \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The two components at the inflow are therefore $u_1^{inf} = c_{Ca^{2+}} - c_{SO_4^{2-}}$ and $u_2^{inf} = c_{Cl^-}$, with the same expressions as in the domain. This contrasts with problems where additional inflow-only equilibria are active (e.g. atmospheric gas equilibria), in which case $\mathbf{U}^{inf}$ would have fewer rows than $\mathbf{U}^{dom}$.

Note on the component matrix used in the simulation. The reactive transport simulation uses the domain component matrix $\mathbf{U}^{dom}$. The component concentrations $\mathbf{u} = \mathbf{U}^{dom} \mathbf{c}$ used in the reactive transport simulation are always computed with the domain component matrix, regardless of whether the underlying species concentrations $\mathbf{c}$ correspond to the domain (initial or interior) or to the boundary (inflow) water. In this particular problem $\mathbf{U}^{inf} = \mathbf{U}^{dom}$, so the projection $\mathbf{u}_{inf} = \mathbf{U}^{dom} \mathbf{c}_{inf}$ coincides with $\mathbf{U}^{inf} \mathbf{c}_{inf}$; nevertheless, the rule is applied consistently with related example problems (e.g. the denitrification and calcite-equilibrium documents) where the two matrices differ.

The initial and boundary concentrations we consider are in Table 1. The initial water in the column is a dilute $CaCl_2$ solution in equilibrium with gypsum, except in one interior cell (cell 11 out of 30) where a high-concentration $CaCl_2$ pulse is imposed (equivalent to dissolved $CaCl_2$ saturated with gypsum). The inflow water has the same chemistry as the resident water. All three waters are charge-balanced and simultaneously satisfy the gypsum equilibrium constraint.

	initial (resident)	pulse (cell 11)	inflow (boundary)
Ca^{2+}	1.0000329	200	1.0000329
Cl^-	2	400	2
$SO_4^{2-}{}^\dagger$	$3.29371 \cdot 10^{-5}$	$1.64691 \cdot 10^{-7}$	$3.29371 \cdot 10^{-5}$

Table 1: Primary species concentrations of initial, pulse, and boundary waters (in mol/kg_w). All three waters are pure $CaCl_2$ solutions saturated with gypsum. [†]Under the ideal-activity assumption ($\gamma_i = 1$, $a_i = c_i$), the SO_4^{2-} concentrations are imposed by equilibrium with gypsum, $c_{Ca^{2+}} c_{SO_4^{2-}} = 10^{-4.4823} \approx 3.2937 \cdot 10^{-5}$. The Cl^- concentrations are set so that each water is electroneutral, $2c_{Ca^{2+}} = c_{Cl^-} + 2c_{SO_4^{2-}}$. Note that in the high- Ca^{2+} pulse cell, the gypsum equilibrium drives SO_4^{2-} to a much lower value (common-ion effect), while in the low- Ca^{2+} resident and inflow water SO_4^{2-} is higher.

64 In terms of components, this translates into

$$\mathbf{u}_{init} = \begin{pmatrix} 1.00000 \\ 2.00000 \end{pmatrix}, \quad \mathbf{u}_{pulse} = \begin{pmatrix} 200.00000 \\ 400.00000 \end{pmatrix}, \quad \mathbf{u}_{inf} = \begin{pmatrix} 1.00000 \\ 2.00000 \end{pmatrix},$$

65 i.e. the components in the resident and inflow water are identical (both contain a dilute $CaCl_2$ solution in equilibrium
66 with gypsum), while the pulse cell has component concentrations roughly two hundred times larger.

67 1.2 Mineral mass balance

68 In addition to the aqueous mass balance (2.1), the gypsum mineral mass per unit volume of porous medium, m_{gyp} ,
69 evolves only through dissolution/precipitation (no transport, since the mineral is immobile):

$$\frac{\partial m_{gyp}}{\partial t} = -r_e, \quad (1.2)$$

70 where r_e is the rate of the gypsum dissolution/precipitation reaction as written in Section 1 (positive when gypsum
71 dissolves, producing aqueous Ca^{2+} and SO_4^{2-}). The initial mineral content corresponds to a volume fraction of 0.50
72 throughout the column, ensuring that gypsum is present everywhere and the equilibrium constraint can be enforced
73 locally at all times during the simulation.

74 2 Transport

We solve the following set of equations:

$$\begin{cases} \phi \frac{\partial \mathbf{c}^T}{\partial t} = \mathcal{L}^T(\mathbf{c}) + \phi \mathbf{r}^T \\ \mathbf{S}_e \log \mathbf{c} = \log \mathbf{K}_e \\ \mathbf{r}_K = \mathbf{r}_K(\mathbf{c}) \end{cases} \quad (2.1)$$

75 where $\phi \in (0, 1]$ is porosity; $\mathbf{c} \in \mathbb{R}^{n_s}$ is the concentration array of all species (n_s is the number of species); $\mathcal{L}^T(\mathbf{c}) =$
76 $-\mathbf{q}^T \nabla \mathbf{c}^T + \nabla \cdot (\mathbf{D} \nabla \mathbf{c}^T) + f_w (\mathbf{c}_r^T - \mathbf{c}^T)$ is the transposed advective form of the transport operator, if we solve transport
77 using the advective ADE, where $\mathbf{q} \in \mathbb{R}^d$ is the Darcy flux (d is the Euclidean dimension, in this case $d = 1$), $\mathbf{D} \in \mathbb{R}^{d \times d}$ is
78 the hydrodynamic dispersion tensor, f_w is the (volumetric) water sink-source, with concentration $\mathbf{c}_r$ if $f_w > 0$ (recharge),
79 or, $\mathbf{c}_r = \mathbf{c}$ if $f_w < 0$ (discharge), or $\mathbf{c}_r = 0$ when $f_w < 0$ represents evaporation; $\mathbf{r} = \mathbf{S}_e^T \mathbf{r}_e + \mathbf{S}_K^T \mathbf{r}_K$ is the contribution of
80 chemical reactions to the mass balance of species (per unit time); $\mathbf{S}_e \in \mathbb{R}^{n_e \times n_s}$ and $\mathbf{S}_K \in \mathbb{R}^{n_K \times n_s}$ are the stoichiometric
81 matrices of equilibrium and kinetic reactions, respectively, whose respective reaction rates are given by $\mathbf{r}_e \in \mathbb{R}^{n_e}$ and
82 $\mathbf{r}_K \in \mathbb{R}^{n_K}$ (n_e and n_K are the number of equilibrium and kinetic reactions, respectively) and $\mathbf{K}_e \in \mathbb{R}^{n_e}$ are the
83 equilibrium constants. In this problem, $n_e = 1$ (gypsum equilibrium) and $n_K = 0$, so $\mathbf{r} = \mathbf{S}_e^T \mathbf{r}_e$.

Equation (2.1) needs to be solved subject to appropriate initial ($\mathbf{c}(\mathbf{x}, t) = \mathbf{c}(\mathbf{x}, 0)$ throughout the model domain,
with a high-concentration $CaCl_2$ pulse imposed in one interior cell) and boundary conditions, which we choose to be
prescribed mass flux, that is,

$$\begin{cases} \mathbf{n}^T (\mathbf{q} \mathbf{u} - \mathbf{D} \nabla \mathbf{u}) = q_{inf} \mathbf{u}_{inf} & \text{at the inflow} \\ \mathbf{n}^T (-\mathbf{D} \nabla \mathbf{u}) = 0 & \text{at the outflow} \end{cases}$$

84 expressing that the flux of components at the inflow is $q_{inf} \mathbf{u}_{inf}$, and that there is no dispersive flux at the outflow. In the
85 component-based formulation actually solved by the simulation, transport is integrated on the conservative components
86 $\mathbf{u} = \mathbf{U}^{dom} \mathbf{c}$ rather than directly on the species concentrations $\mathbf{c}$, which removes the equilibrium reaction term from the
87 transport step.

88 To solve numerically, we choose a uniform mesh made of 30 cells of length $\Delta x = 0.1$ m (domain length $L = 3$ m).
89 The pulse is imposed at cell 11. The time step is $\Delta t = 10^{-2}$ d, and the simulation is run for a total time of $T = 10$ d
90 (i.e. 1000 time steps), long enough for the pulse to be advected and dispersed across most of the column at the chosen
91 Darcy velocity ($qT/\phi = 4$ m $> L$). We use the fully implicit Euler scheme for both transport and reactions. We consider
92 uniform transport properties: $\phi = 0.5$, $q = q_{inf} = 0.2$ m/d, $\alpha_L = 1$ m. The dispersion coefficient is $D = \alpha_L q = 0.2$ m²/d.
93 Note that while the chemistry described in Section 1 (species, gypsum reaction, and equilibrium constant) is intrinsic
94 to this gypsum equilibrium problem, the transport parameters and numerical settings — mesh size, time step, porosity,
95 Darcy flux, dispersivity, boundary condition type, and integration scheme — can be freely modified by the user to suit
96 different physical scenarios, subject to the following stability and accuracy constraints:

- **Grid Péclet number:** $Pe = \frac{q\Delta x}{D} = \frac{\Delta x}{\alpha_L} \leq 2$ to avoid spurious oscillations in centred schemes. In this example, $Pe = 0.1$.
- **Courant number:** $Co = \frac{q\Delta t}{\phi\Delta x} \leq 1$ for time integration. In this example, $Co = 0.2 \cdot 10^{-2} / (0.5 \cdot 0.1) = 0.04$.
- **Von Neumann stability:** For the explicit Euler scheme, the time step must satisfy $\Delta t \leq \frac{\phi(\Delta x)^2}{2D}$ for dispersion stability. The fully implicit Euler scheme is unconditionally stable, so this constraint does not apply.
- **Combined constraint (explicit Euler):** When using explicit time integration, both the Courant and dispersion constraints must be satisfied simultaneously: $\Delta t \leq \min\left(\frac{\phi\Delta x}{q}, \frac{\phi(\Delta x)^2}{2D}\right)$.

The fully implicit Euler scheme used in this example is unconditionally stable, allowing larger time steps at the cost of solving a nonlinear system at each step. However, accuracy still requires Δt small enough to resolve the dissolution/precipitation dynamics induced by the pulse.

References

- De Simoni, M., Carrera, J., Sanchez-Vila, X., and Guadagnini, A. (2005). A procedure for the solution of multicomponent reactive transport problems. *Water resources research*, 41(11).